Process migration on a RISC-V microkernel

An empirical comparison of migration call paths

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Why revisit migration placement?

Research question

How much migration cost does each call path add on the same Hydra configuration?

1993 Mach reported a substantial user-space penalty under its test conditions.

Reasons for new measurements

IPC optimisation

Communication mechanisms have changed.

Deployment demand

State preservation has practical use cases.

RISC-V adoption

An open baseline should be measured on the chosen ISA.

Report §§1.2, 1.6, 2.3.2. Milojičić et al. (1993, 1999), Gu et al. (2020), Chen et al. (2024).

Five research objectives

01	**Implement**	Compare direct and wrapper migration call paths.
02	**Measure**	Record intervals, kernel-code contributions and calls.
03	**Compare**	Assess performance and shared kernel requirements.
04	**Reassess**	Revisit the 1993 finding and 1999 adoption barriers.
05	**Reproduce**	Provide source code, raw data and runnable benchmarks.

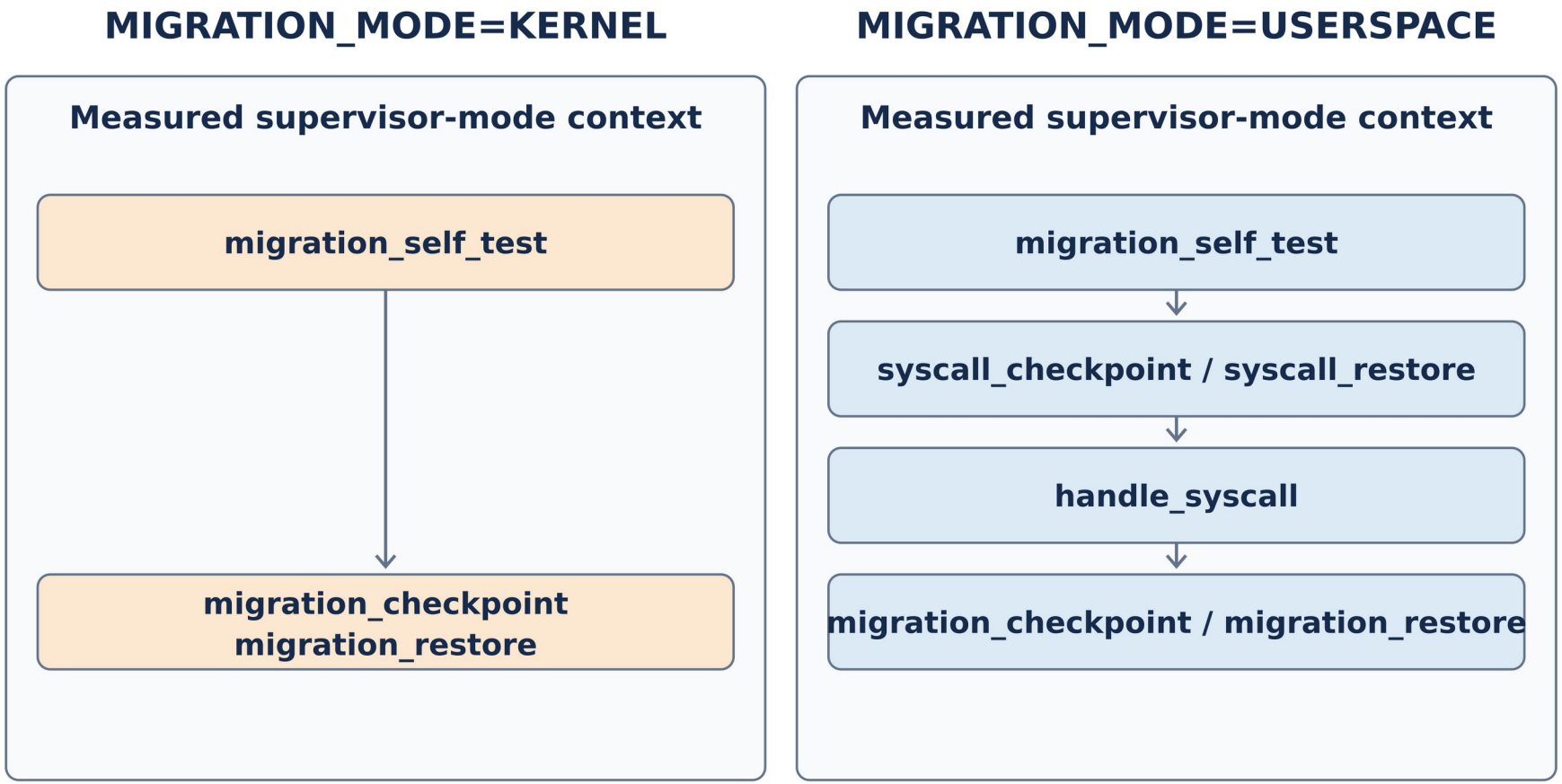
Report §1.3. Research objectives.

How the implementation developed

1	**Hydra scaffold**	Inherited the teaching kernel and application structure.
2	**Migration mechanism**	Added context/stack primitives. 393 lines in migration.c and migration.h.
3	**Repeatable execution**	Fixed task-slot reuse and the frozen startup counter.
4	**Measurement harness**	Added mode selection, BENCH timestamps and analysis.
5	**Later functional work**	Added ACK/retry, isolated images and verified demo inputs.

Report §§5.2.1–5.2.9. Repository contribution log: docs/changes.md.

Both paths share checkpoint and restore



Both paths call the same primitives; no hardware-enforced user/kernel transition is measured.

Report Fig. 5.3, §§5.2.1, 5.2.4. Both historical paths execute in supervisor mode.

Controlled experimental design

120

local observations

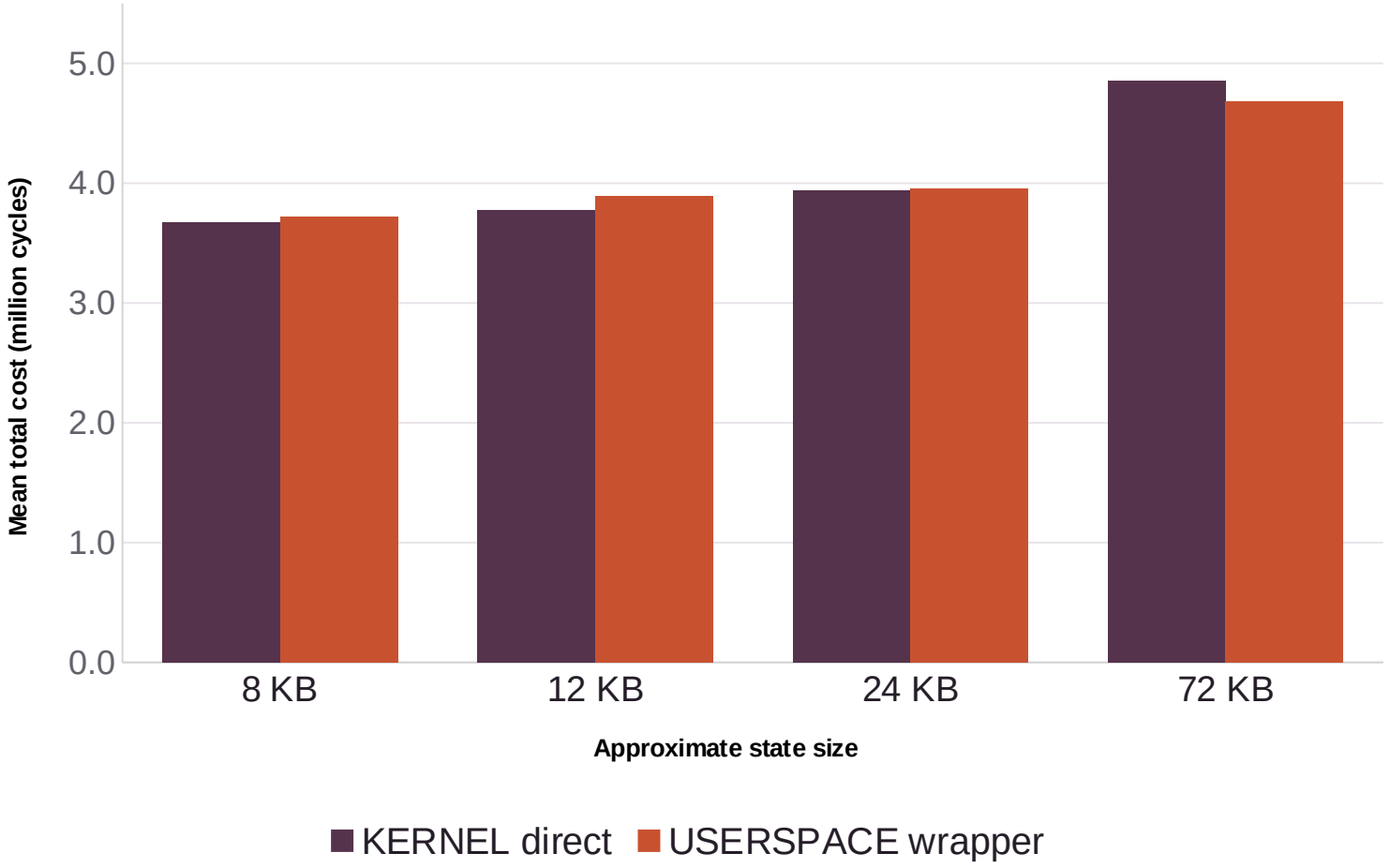
2 modes × 4 sizes
× 15 repetitions

Control	Configuration
Platform	Same Hydra and QEMU RISC-V64 virt
Workload	Counter task and fixed 8 KB stack
Extra payload	0, 4, 16 and 64 KB
Instrumentation	9 timestamps defining 5 intervals
Analysis	Means and separate 95% confidence intervals

Approximate state sizes: 8, 12, 24 and 72 KB. Extra bytes are synthetic.

Report Chapter 3, §§4.2.1–4.2.4. Experimental unit: one migration run.

Similar means under the measured conditions



1.00

pooled wrapper/direct ratio
rounded to two decimals

4,061,687

KERNEL mean cycles

4,061,801

USERSPACE mean cycles

15 observations per size and mode. Similar means do not establish equivalence.

Original dataset: bench/results/KERNEL.log and USERSPACE.log. Report §§4.2.1–4.2.4.

Checkpoint and restore produce a running task

8,784 bytes
Checkpoint captured

166 / 167
Original / restored PID

+20
Counter delta

PASS
Restored task running

```
[migtest] checkpoint OK, 8784 bytes written
BENCH send_start 414673377401540
BENCH send_done 414673377907553
BENCH recv_start 414673378412946
BENCH recv_done 414673378966451
BENCH rest_start 414673381011242
[migration] restore: old_pid=166 new_pid=167 sp_off=8112
BENCH rest_done 414673382562668
[migtest] restore OK, new pid=167
[migtest] after restore, counter=2370 (delta=20)
[migtest] PASS: restored task is running
[sched] Terminated task 166
[sched] Terminated task 167
[migtest] Local migration self-test complete
```

Optional 36-second QEMU demonstration

Later QEMU recording, 14 Oct 2026. Actual UART output with a recreated terminal appearance.

Transport reliability required a second iteration

8 / 15

Original completions

Best-effort multicast
No ACK or retransmission

15 / 15

Later supervisor restores

ACK/retry validation
Plus 1 isolated-task restore

Current acceptance audit

```
supervisor=15/15  isolated=1/1  acks=16/16  errors=0
```

Report §4.2.5, §6.3.2. Original sample and later acceptance logs are separate.

The adoption barriers depend on deployment context

1999 barrier	Reassessment in the report
Lack of applications	Weaker in selected state-preservation use cases
Lack of infrastructure	Weaker, with platform support still varying
Simpler alternatives	Depends on whether execution state must persist
Sociological resistance	Depends on ownership, tenancy and policy

Contextual interpretation alongside the measurements. No universal architectural verdict.

Report §6.2.2 and Fig. 6.1. Milojičić et al. (1999, §7.3).

Limitations and next measurements

Experimental limit	Required follow-on measurement
Supervisor-mode comparison	Measure real protected-server transitions and IPC
QEMU execution	Repeat on calibrated physical RISC-V hardware
Bounded process state	Test private heaps and external resource ownership
Best-effort original transport	Measure retries, loss, failed attempts and tail latency

Production suitability and statistical equivalence remain questions for further experiments.

Report §§1.7, 4.2.10, 6.2.4 and 6.3. Later hardening is separate evidence.

What the project established

Small observed wrapper cost in the measured baseline

Objective	Evidence retained
Implement	Two mode builds and shared checkpoint/restore primitives
Measure	BENCH intervals, source calls and contribution records
Compare	60 observations per mode and matched-size analysis
Reassess	Cited analysis of historical findings and barriers
Reproduce	Source, raw logs, test scripts and capture metadata

github.com/sahanruwantha/Hydra-Microkernal

Report abstract and Chapter 6. Repository, retained datasets and demo evidence archive.

Backup: phase means and uncertainty

Metric (cycles)	KERNEL mean ± 95% half-width	USERSPACE mean ± 95% half-width
Freeze	275,362 ± 9,117	272,834 ± 7,245
Serialize	1,078,726 ± 55,295	1,076,715 ± 48,517
Local transport tags	782,891 ± 15,123	785,910 ± 14,762
Deserialize	1,029,199 ± 70,017	1,016,759 ± 59,260
Total to READY	4,061,687 ± 145,479	4,061,801 ± 118,173

Migration primitives: 319 C lines + 74 header lines = 393 lines

bench/results/comparison.txt. Report Tables 4.1–4.2 and §§4.2.7, 5.2.1.

Backup: controlled task recovery

Frozen counter: 59

After recovery: 88 (+29)

```
[fdemo] Step 3: checkpoint saved (8784 bytes) at counter=30
[sched] Resumed task 168
[fdemo] Step 4: workload progressed past checkpoint, counter=59
[fdemo] Step 5: SIMULATING FAULT — killing task 168
BENCH fdemo_fault_inject 414695554900244
[sched] Terminated task 168
[fdemo] Step 6: detected failure (counter frozen at 59)
BENCH fdemo_recovery_start 414695595814323
BENCH rest_start 414695596497284
[migration] restore: old_pid=168 new_pid=169 sp_off=8112
BENCH rest_done 414695598069604
BENCH fdemo_recovery_done 414695598557420
[fdemo] Step 7: restored from checkpoint as pid=169
[fdemo] Step 8: post-recovery counter=88 (delta from fault=29)
[fdemo] PASS: workload recovered from simulated fault
[sched] Terminated task 169
[fdemo] === Demonstration complete ===
```

Later QEMU r-command recording. Report §4.2.6 describes the bounded task-recovery demonstration.

Selected references

Milojičić et al. (1993) Task Migration on Top of the Mach Microkernel. USENIX Mach Symposium, pp. 273–290.

Milojičić et al. (1999) Process Migration. HP Laboratories Technical Report HPL-1999-21.

Gu et al. (2020) Harmonizing Performance and Isolation in Microkernels with Efficient Intra-kernel Isolation and Communication. USENIX ATC, pp. 401–417.

Chen et al. (2024) Microkernel Goes General: Performance and Compatibility in the HongMeng Production Microkernel. USENIX OSDI, pp. 465–485.

Clark et al. (2005) Live Migration of Virtual Machines. USENIX NSDI, pp. 273–286.

Waterman and Asanović (2019) The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA. Version 20191213.

Full Harvard reference list: Final_Report.docx, References. Figure/data sources appear in slide notes.